

Manuel Mezo

Madrid, Spain | +34 652257366 | manumezog@gmail.com

[Github](#) | [Project portfolio](#) | [LinkedIn](#)

PROFILE

Engineer and applied AI builder with an **8+ year track record** of solving **multi-million-dollar operational challenges** at Amazon and McKinsey, now independently **shipping production AI systems**: end-to-end RAG pipelines, autonomous LLM agents, computer vision, and multi-agent simulations. Combines deep operational instinct for what AI needs to do for real users with hands-on engineering capability across the full stack: Python, FastAPI, cloud infrastructure, ML fine-tuning, and real-time deployment. Seeking an **AI Engineer, Technical Product Manager (TPM), or Product Manager (PM)** role in a high-growth startup to bridge the gap between cutting-edge AI capabilities and scalable business solutions.

TECHNICAL SKILLS

AI & LLM: LangChain, End-to-End RAG pipelines, **Chroma Vector DB**, **LLM APIs (Claude, Gemini 2.5, GPT-4)**, multi-agent systems, prompt engineering, structured outputs, voice agent stacks (**Retell.ai**, **Voiceflow**). **Computer Vision & ML:** **PyTorch**, **Weights & Biases (W&B)**, **OpenCV**, **YOLOv8** fine-tuning (~97% mAP@50), **ByteTrack**, **Ultralytics**, **Roboflow**. **Backend & Cloud:** **FastAPI**, **Next.js**, **Vercel** serverless functions, **GCP**, **AWS**, **Docker**, **Firebase**, **Neon Postgres**, **BigQuery**. **Development & Data:** **Python**, **TypeScript**, **C++**, **SQL**, **Git/GitHub**, **CI/CD**, **Pandas**, **NumPy**, **Tableau**.

APPLIED AI PROJECTS

- **Portfolio Chat Agent — End-to-End RAG + FastAPI** ([link](#)): Architected a production-ready conversational AI agent from scratch using **LangChain**, **ChromaDB**, and **FastAPI**. Overhauled knowledge base for optimal retrieval via semantic story-level chunking. Containerized and deployed the backend via **Docker** to Hugging Face Spaces with robust conversational memory and rate limiting. Integrated traces with **Langfuse** for observability.
- **FastPay AI — Autonomous 3-Way Invoice Matching Agent** ([link](#)): Built a full-stack agentic pipeline (**Next.js**, **TypeScript**, **Neon Postgres**) that extracts data from messy real vendor pdf invoices using **Gemini 2.5 Flash**, cross-references POs/Receipts data, and autonomously approves or flags discrepancies. Engineered 7 custom tools, deterministic decision rules, a **real-time live tracing UI** and an **evaluation** suite utilizing **Langfuse** observability.
- **AI Voice Customer Service Agent** ([link](#)): Deployed a production voice agent using **Retell.ai** and **Voiceflow**, integrated with real-time order lookup via **Vercel serverless functions**. Designed for low-latency multi-turn conversation with live backend data retrieval.

COMPUTER VISION AND ROBOTICS PROJECTS

- **Custom YOLOv8 Object Tracking & Spatial Logic** ([link](#)): Engineered a custom detection pipeline using an annotated dataset in **Roboflow**. Fine-tuned a **YOLOv8n model** (~97% mAP@50) and implemented real-time tracking via **ByteTrack** and **OpenCV**. Developed advanced spatial logic for moving conveyor belts featuring point-in-polygon zones.
- **Integrated VLA AI model in 3D Printed Robotic Arm** ([link](#)): Built a physical 4-axis robotic arm integrated with a cloud-based Vision-Language-Action (VLA) reasoning engine. The system utilizes **Gemini 1.5 Pro** to translate natural language voice commands into precise mechanical actuations.

PROFESSIONAL EXPERIENCE

Manager, EU Transportation Forecasting | Amazon (Madrid, Spain)

Feb 2026 – Present

- Lead the technical roadmap for EU transportation forecasting, targeting **>\$10M/year in savings** through **ML bias** correction, automated **simulation** modules, and data **pipeline health** checks.
- Own **end-to-end production** of weekly demand forecasts covering all EU logistics network segments, driving transportation and labor capacity **optimization**.
- Manage a distributed team of 6 Business Intelligence Engineers and Program Managers; lead **roadmap prioritization** with the Forecasting Tech engineering team.

Sr Program Manager, Supply Chain Planning | Amazon (Madrid, Spain)

Sept 2023 – Feb 2026

- **Invented** and implemented a long-range (52-week) forecasting methodology for EU truckload demand, supporting S&OP cycles and driving **\$140Mn CapEx decisions** for 2025.
- Improved inventory transfer truck fill rate modeling (**inaccuracy reduced from 35% to 17%**), **saving ~\$1.8Mn annually** by preventing late additions and cancellations.
- Identified and resolved a **critical logic defect** in the automated scheduling system using SQL-based data analysis and Python scripting, **protecting ~\$10M** in avoidable late truck addition costs.

Operations Consultant | McKinsey & Company (Madrid, Spain)

Aug 2022 – Jul 2023

- **Redesigned** and standardized the production capacity planning tool of a global pharmaceutical manufacturer through an **automated**, KPI-oriented model and **rolled it out** in 6 plants (13 production lines).
- Optimized the production plan and timelines of a train manufacturer, recovering 2.5 months of delivery delays worth **\$1M in saved penalties**.

Sr Program Manager, 3PL Ops Excellence & Quality | Amazon (Luxembourg)

Sept 2017 – Aug 2022

- Led the planning and execution of multi-item order shipping capabilities across 10 3PL sites, generating **\$10Mn+ in transport savings**.
- Designed network-wide inventory audit programs across 25+ warehouses, **improving the Inventory Record Defect Rate metric by 50%**.
- Recovered a severe inventory defects crisis from **120k to <10k DPMO in 8 weeks**, driving end-to-end process redesign.

EDUCATION

- **MSc in Aerospace Engineering** | Universidad Carlos III de Madrid & TU Delft (Exchange)
- **BSc in Aerospace Engineering & BSc in Mechanical Engineering** | Universidad de León
- [Python for data science certification](#) by Microsoft (2016)

LANGUAGES

- **Spanish — Native | English — Proficient (C1) | French — Intermediate (B2)**